

Set-up: Let B be a topological space.

Consider outcome relations of the following type:

$$pR\left(\bigcup_{\alpha} T_{\alpha}\right), T_{\alpha} \text{ closed} \Rightarrow pR\left(\bigcup_{\alpha} T_{\alpha}\right) \cup \left(\bigcup_{\beta} T_{\beta}\right)$$

where T_{β} is closed.

Union of closed sets can be put inside a ~~new~~ bigger union of closed sets. This is far from full blown monotonicity.

Algebra: Concrete and Abstract: Form Algebra with pairs (R, S) where R, S are outcome relations on a topological space of the above kind.

Definition of $\cup, \cap, -$ remain unchanged.

Introduce a new operator called topo-composition as follows:

$$R \circ S := \left\{ (p, T) \mid (p, U) \in R \text{ where } U = \bigcup_{\alpha} U_{\alpha}, \right. \\ \left. U_{\alpha} \text{ closed}, (u, T) \in S \text{ for all } u \in U \right\}.$$

Call this class of concrete algebras topo-algebra.

The axiom G3 needs checking that's all:

Must try to show: $R_1 \circ (R_2 \circ R_3) \subseteq (R_1 \circ R_2) \circ R_3$:

Let $(p, T) \in R_1 \circ (R_2 \circ R_3)$

$\therefore (p, U) \in R_1$ $U = \bigcup_{\alpha} U_{\alpha}$; U_{α} closed, $(u, T) \in R_2 \circ R_3 \forall u \in U$.

$\forall u \in U, \exists T_u = \bigcup_{\beta} T_{\beta u}$; $T_{\beta u}$ closed such that

$(u, T_u) \in R_2, (t', T) \in R_3 \forall t' \in T_u$.

$\bigcup_{u \in U} T_u = \bigcup_{u \in U} \bigcup_{\beta} T_{\beta u}$: union of closed sets

$\therefore (u, \bigcup_{u \in U} T_u) \in R_2$ and $(t', T) \in R_3 \forall t' \in \bigcup_{u \in U} T_u$

$$\therefore (P, \bigcup_{u \in U} T_u) \in R_1 \circ R_2$$

$$(t', T) \in R_3 \quad \forall t' \in \bigcup_{u \in U} T_u$$

$$\therefore (P, T) \in (R_1 \circ R_2) \circ R_3$$

GOOD FOR US !!

So, Topo-Algebra is a game algebra.

Toolkit: Game Module, Filter, Prime Filter,
~~undergo no change~~ $\hat{A}$, F_T undergo no change.

The reader can verify that $\hat{A}$'s form basic closed sets in the space of prime filters.

CONSTRUCTION OF OUTCOME RELATION
ON THE ~~BOARDS~~ PRIME FILTER SPACE:

Let $D = (D, \vee, \wedge)$ be a distributive lattice
and Δ be a monotone map.

Follow the steps below to construct the suitable
kind of outcome relation on the prime filter space $\mathcal{B}_D$.

$$\textcircled{1} \quad A = \{ \bigcup T_\alpha \mid T_\alpha \text{ closed} \}$$

$$A = \{ \emptyset \mid B = \bigcup_{\substack{\alpha \\ \in I}} T_\alpha \text{ for some indexing set } I \\ \text{such that } T_\alpha \text{ is closed} \}.$$

$$A = \{ C \in A \mid C \text{ is closed} \} \cup \{ C \in A \mid C \text{ is not closed} \}.$$

Let $A_1 := \{C \in A \mid C \text{ is closed}\}$

$A_2 := \{C \in A \mid C \text{ is not closed}\}.$

We will define Q_Δ on ~~power~~ sets of A_1, A_2 .

$C \in A_1$: C is closed.
 $p Q_\Delta C$ iff $\Delta a \in p$ for all $a \in F_C$.

$C \in A_2$: ~~C is not closed.~~ C is not closed.

~~$p Q_\Delta C$ iff~~ $\therefore C = \bigcup_{\alpha \in I} T_\alpha$ T_α closed, for some indexing set I .

$p Q_\Delta C$ iff $\exists T$ closed, $T \subseteq C$ and $p Q_\Delta T$.

checking this satisfies the restricted monotonicity:

Let $pQ_{\Delta} \bigcup_{\alpha \in I} T_{\alpha}$; where T_{α} is closed.

Want to show: $pQ_{\Delta} \left(\bigcup_{\beta \in J} T_{\beta} \right) \cup \left(\bigcup_{\alpha \in I} T_{\alpha} \right)$

Case 1: $\bigcup_{\alpha \in I} T_{\alpha}$ is closed.

Let $T = \bigcup_{\alpha \in I} T_{\alpha}$.

Let $U = \left(\bigcup_{\beta \in J} T_{\beta} \right) \cup \left(\bigcup_{\alpha \in I} T_{\alpha} \right)$

U closed

$T \subseteq U \Rightarrow F_U \subseteq F_T$

$\Rightarrow \Delta_a p \in F_U$

for all $a \in F_U$.

$\therefore pQ_{\Delta} U$.

U not closed

$T \subseteq U$, T closed.

and $pQ_{\Delta} T$.

$\therefore pQ_{\Delta} U$.

Case 2: $\bigcup_{\alpha \in I} T_{\alpha}$ is not closed.

$\therefore pQ_{\Delta} T$ for some closed set $T \subseteq \bigcup_{\alpha \in I} T_{\alpha}$.

$\therefore T \subseteq \left(\bigcup_{\beta \in J} T_{\beta} \right) \cup \left(\bigcup_{\alpha \in I} T_{\alpha} \right)$

The rest follow from Case 1.

This outcome relation isn't monotone on the full space since sets may not be in A_1 and A_2 .

This indeed is restricted form of monotonicity.

Key ~~Propositions~~ Lemmas:

① $p \leq_Q \hat{a}$ iff $\Delta a \in p$.

Proof: $\Rightarrow$: $\hat{a}$ is closed (basic closed set)

$\therefore p \leq_Q \hat{a}$ iff $\Delta a \in p$ since $a \in F_{\hat{a}}$.

$\Leftarrow$: Let $\Delta a \in p$.

Since $\hat{a}$ is closed, one must show

$\Delta b \in p$ when $b \in F_{\hat{a}}$.

Let $b \in F_{\hat{a}}$.

$\therefore \hat{a} \subseteq \hat{b}$.

$\therefore a \leq b$.

$\therefore \Delta a \leq \Delta b$.

$\therefore \Delta b \in p$ (by upward closure of p)

② For any distributive lattice D , the map $\hat{(\cdot)}$ is an embedding of D in $(P(BD), \cup, \cap)$.

Proof: No change.

③ Prime Filter Theorem:

Main Propositions:

① Let $M = (M, \vee, \wedge, -, \Delta_g)_{g \in G}$ be a module over the game algebra $\mathcal{G} = (G, \vee, \wedge, -, \Delta)$, and let g and h be arbitrary elements of G . Then:

1. $\mathcal{Q}_{g \vee h} = \mathcal{Q}_g \cup \mathcal{Q}_h$

2. $\mathcal{Q}_{g \wedge h} = \mathcal{Q}_g \cap \mathcal{Q}_h$

3. $\mathcal{Q}_{g \Delta h} = \mathcal{Q}_g \Delta \mathcal{Q}_h$

4. If $\Delta_g a \neq \Delta_h a$ for some $a \in M$, then $\mathcal{Q}_g \neq \mathcal{Q}_h$.

Before discussing the proof, let us see how this ~~relation~~ ~~the~~ ~~homom~~ gives rise to homomorphism:

② Let $M = (M, \vee, \wedge, -, \Delta_g)_{g \in G}$ be a separable module over the game algebra $\mathcal{G} = (G, \vee, \wedge, -, \Delta)$. Then the map $\text{rep}: G \rightarrow G(BM)$ given by

$$\text{rep}(g) = (\mathcal{Q}_g, \mathcal{Q}_{-g}),$$

is an embedding of $\mathcal{G}$ into $\mathcal{G}_m(BM)$.

Proof: Immediate from ① and separability of the module.

③ Let $g = (G, V, 1, -, \Delta)$ be a game algebra.

Then g , seen as a game module over itself, can be embedded in a separable game module g' over g .

Proof: As before.

Let us turn to the proof of ①.

No changes in 1, 2, 4.

Let us prove 3.

Suppose that $p(Q_g \circ Q_h)T$ ~~is not~~ closed.

$\exists U = \bigcup_{\alpha} U_{\alpha}$; U_{α} closed such that $pQ_g U$ and

$uQ_h T$ for all $u \in U$.

Suppose T is closed.

Let a be such that $T \subseteq a^{\wedge}$.

By definition, $\Delta_h a \in U$ for all $a \in F_T$.

$\therefore U \subseteq (\Delta_h a)^{\wedge}$

If U is closed, then $pQ_g U \Rightarrow pQ_g (\Delta_h a)^{\wedge}$

If U is not closed, $\exists \mathcal{C} \subseteq U$ closed such that

$\mathcal{C} \subseteq U$ and $pQ_g \mathcal{C}$

So $\mathcal{C} \subseteq (\Delta_h a)^{\wedge}$.

$\therefore pQ_g (\Delta_h a)^{\wedge}$

$\therefore \Delta_g \Delta_h a \in p$

We have $\Delta_g \Delta_h a = \Delta_{g \circ h} a$.

$\therefore \Delta_{g \circ h} a \in p$.

So, $p(Q_{g \circ h})T$.

$\begin{aligned} & C \subseteq T, \Delta_h C \subseteq U, \Delta_g \Delta_h C \subseteq p \\ & \Delta_h C \subseteq T, \Delta_g \Delta_h C \subseteq p \\ & \Delta_h C \subseteq T, \Delta_g \Delta_h C \subseteq p \\ & \Delta_h C \subseteq T, \Delta_g \Delta_h C \subseteq p \end{aligned}$

Not done: When T is not closed.

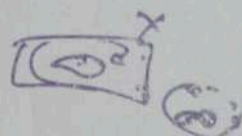
For the other direction, suppose $p \mathcal{Q}_g \cap T$.

First assume T is closed.

$$U = \cap \{ (\Delta_h b)^\wedge \mid b \in F_c \}.$$

$$\forall u \in U, b \in F_c, \Delta_h b \in u.$$

$$\therefore u \mathcal{Q}_h \subset \text{ for every } u \in U.$$



~~Fractured~~.

$A = \{ \Delta_h b \mid b \in F_c \}$ is downward directed.

By lemma, it suffices to show, $\Delta_g a \in p$ for every $a \in A$ to prove $p \mathcal{Q}_g U$.

Take $a \in A$. It is of the form $\Delta_h b$ for some $b \in F_c$.

$$p \mathcal{Q}_{g \Delta h} \subset \Rightarrow \Delta_{g \Delta h} b \in p.$$

$$\therefore \Delta_{g \Delta h} b = \Delta_g \Delta_h b = \Delta_g a \in p.$$

~~In case T is not closed, we may use the above proof to infer $p \mathcal{Q}_{g \Delta h} \subset$ when $c \in T$ and $p \mathcal{Q}_h \subset$.~~

Not done: T is closed.